

Proof of Lemma 0.1.3.

Let $P^* \geq P$ be a refinement of P . Since partition contains only finite elements,

WLOG, assume that $P^* = P \cup \{S\}$ s.t. $S \notin P = \{x_0, \dots, x_n\}$

Then, for some $1 \leq j \leq n$ $x_{j-1} < S < x_j$. Now,

$$\begin{aligned} U(f, P, \alpha) - U(f, P^*, \alpha) &= \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(x_j) - \alpha(S)] + \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(S) - \alpha(x_{j-1})] \\ &\quad - \sup_{t \in [x_{j-1}, S]} f(t) \cdot [\alpha(x_j) - \alpha(S)] - \sup_{t \in [S, x_j]} f(t) \cdot [\alpha(S) - \alpha(x_{j-1})] \\ &\geq 0. \quad \square \end{aligned}$$

Proof of Lemma 0.1.4. X fail try.

Since $f, g \in R(\alpha)$, $\int_a^b f \, d\alpha = \int_a^b f \, d\alpha$ and $\int_a^b g \, d\alpha = \int_a^b g \, d\alpha$. Now,

$$\begin{aligned} \int_a^b f + g \, d\alpha &= \inf_{P \in \mathcal{P}_{[a,b]}} U(f+g, P, \alpha) = \inf_{P \in \mathcal{P}_{[a,b]}} \left[\sum_{i=1}^n M_i(f+g) \cdot \Delta\alpha_i \right] \\ &\leq \inf_{P \in \mathcal{P}_{[a,b]}} \left[\sum_{i=1}^n M_i(f) \cdot \Delta\alpha_i + \sum_{i=1}^n M_i(g) \cdot \Delta\alpha_i \right] \end{aligned}$$

Proof of Thm 0.1.4

1) $\Rightarrow$ 2). Since $f \in \mathcal{R}(a)$, given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ s.t.

$$\int_a^b f \, d\alpha \leq U(f, P, \alpha) < \int_a^b f \, d\alpha + \frac{\varepsilon}{2} \quad \text{and} \quad \int_a^b f \, d\alpha \geq L(f, P, \alpha) > \int_a^b f \, d\alpha - \frac{\varepsilon}{2}$$

Ask the next step to a toddler on the street.

2) $\Rightarrow$ 1). It is clear, because for any partition $P \in \mathcal{P}[a, b]$,

$$\int_a^b f \, d\alpha - \int_a^b f \, d\alpha \leq U(f, P, \alpha) - L(f, P, \alpha)$$

To show 1) $\Leftrightarrow$ 3), we prove a statement: If 2) holds, then for any $1 \leq i \leq n$,

$$\text{for arbitrary } s_i, t_i \in [x_{i-1}, x_i], \quad \sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i < \varepsilon.$$

$$\text{Since } |f(s_i) - f(t_i)| \leq M_i - m_i,$$

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \cdot \Delta\alpha_i \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

1) $\Rightarrow$ 3). Suppose $f \in \mathcal{R}(a)$. Then, by 2), there exists a partition $P_0 \in \mathcal{P}[a, b]$ s.t.

$$U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Now, for any refinement $P > P_0$,

$$U(f, P, \alpha) - L(f, P, \alpha) \leq U(f, P_0, \alpha) - L(f, P_0, \alpha) < \varepsilon$$

Clearly, regardless of choice of t_i ,

$$L(f, P, \alpha) \leq S(f, P, \alpha) \leq U(f, P, \alpha)$$

$$\text{and} \quad U(f, P, \alpha) \geq \int f \, d\alpha > L(f, P, \alpha)$$

$$\text{Therefore, } \left| S(f, P, \alpha) - \int f \, d\alpha \right| \leq U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

3) $\Rightarrow$ 1) is trivial.

Proof of lemma 0.1.5.

Let $f, g \in \mathcal{R}(a)$ be given. Given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ such that

$$U(f, P, \alpha) - L(f, P, \alpha) < \frac{\varepsilon}{2} \quad \text{and} \quad U(g, P, \alpha) - L(g, P, \alpha) < \frac{\varepsilon}{2}$$

(Technically, take $P_1, P_2 \in \mathcal{P}[a, b]$ for f, g respectively, and put $P = P_1 \cup P_2$). Now,

$$U(f+g, P, \alpha) - L(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) - L(f, P, \alpha) - L(g, P, \alpha) < \varepsilon.$$

And the equality is obtained by: Given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a, b]$ s.t.

$$\begin{aligned} \int_a^b f \, d\alpha + \int_a^b g \, d\alpha - \varepsilon &\leq L(f, P, \alpha) + L(g, P, \alpha) \leq L(f+g, P, \alpha) \\ &\leq \int_a^b f+g \, d\alpha \\ &\leq U(f+g, P, \alpha) \leq U(f, P, \alpha) + U(g, P, \alpha) \leq \int_a^b f \, d\alpha + \int_a^b g \, d\alpha + \varepsilon \end{aligned}$$

The second assertion is clear. Now, we can say that

$\mathcal{R}(a)$ is a $\mathbb{R}$ -vector space.

Proof of lemma 0.1.6.

Suppose that $f \in \mathcal{R}(a)$ on $[a, b]$. Given $\varepsilon > 0$, there exists $P \in \mathcal{P}[a, b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

Put $P^* = P \cup \{c\}$. Then, for $P_1 = P^* \cap [a, c]$ and $P_2 = P^* \cap [c, b]$. Now,

$$\begin{aligned} \int_a^c f \, d\alpha - \int_a^c f \, d\alpha + \int_c^b f \, d\alpha - \int_c^b f \, d\alpha &\leq U(f, P_1, \alpha) - L(f, P_1, \alpha) + U(f, P_2, \alpha) - L(f, P_2, \alpha) \\ &= U(f, P^*, \alpha) - L(f, P^*, \alpha) < \varepsilon \end{aligned}$$

Similarly, the converse holds.

Proof of Thm 0.1.8.

Given $\epsilon > 0$, take $\eta > 0$ s.t. $[\alpha(b) - \alpha(a)] \cdot \eta < \epsilon$.

Since f is continuous on the compact set $[a, b]$, it is continuous uniformly.

Thus, there exists $\delta > 0$ s.t.

$$\forall x, y \in [a, b], |x - y| < \delta \Rightarrow |f(x) - f(y)| < \eta.$$

Now, let $P \in \mathcal{P}[a, b]$ be a partition such that for all i , $\Delta x_i < \delta$ ($P = \{x_0, \dots, x_n\}$)

Then, for all i , $M_i - m_i \leq \eta$. Now,

$$U(f, P, \alpha) - L(f, P, \alpha) = \sum_{i=1}^n (M_i - m_i) \cdot \Delta \alpha_i \leq \sum_{i=1}^n \eta \cdot \Delta \alpha_i = \eta \cdot \sum_{i=1}^n \Delta \alpha_i = \eta \cdot [\alpha(b) - \alpha(a)] < \epsilon.$$

only

Proof of Thm 0.1.9. Alternatively, since monotone function can have ^{only} countably simple discontinuity, if no such shared discontinuities with f and α , then countably discontinuous f is R.S. integrable. But, in this, prove by elementary, using properties of monotone.

Since α is continuous, for each $n \in \mathbb{N}$, there exists a partition $P_n \in \mathcal{P}[a, b]$,

$$\Delta \alpha_i = \frac{\alpha(b) - \alpha(a)}{n} (= \Delta \alpha) \text{ for all } i$$

(More specifically, let $n \in \mathbb{N}$ be fixed. By the Intermediate Value Theorem, there exists $x_1 \in [a, b]$ s.t. $\alpha(x_1) = \alpha(a) + \frac{\alpha(b) - \alpha(a)}{n}$. Inductively,

there exists $x_2 \in [x_1, b]$ s.t. $\alpha(x_2) = \alpha(a) + 2 \cdot \frac{\alpha(b) - \alpha(a)}{n}$

$\vdots$

$x_{n-1} \in [x_{n-2}, b]$ s.t. $\alpha(x_{n-1}) = \alpha(a) + (n-1) \cdot \frac{\alpha(b) - \alpha(a)}{n}$

And take $x_n = b$. Essentially, the existence such partition is obtained by I.V.P.

But, Monotone can have only first-kind discontinuity and

I.V.P. implies it can not have first-kind discontinuity, thus Monotone I.V.P. $\Rightarrow$ Continuity.

WLOG,

Given $\epsilon > 0$, Assume f is monotone increasing. Take $n \in \mathbb{N}$ s.t. $\frac{[\alpha(b) - \alpha(a)] \cdot [f(b) - f(a)]}{n} < \epsilon$.

$$U(f, P_n, \alpha) - L(f, P_n, \alpha) = \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = \sum_{i=1}^n [f(x_i) - f(x_{i-1})] \Delta \alpha_i = \Delta \alpha \cdot \sum_{i=1}^n [f(x_i) - f(x_{i-1})]$$

$$= \Delta \alpha \cdot [f(b) - f(a)] < \epsilon. \quad \square$$

Proof of Thm. 0.1.10.

Since ϕ is continuous on the compact set $[m, M]$, it is continuous uniformly.

Given $\epsilon > 0$, there exists $0 < \delta < \epsilon$ s.t. $|s - t| < \delta \Rightarrow |\phi(s) - \phi(t)| < \epsilon$.

Since $f \in \mathcal{R}(\alpha)$ on $[a, b]$, there is a partition $\{x_0, \dots, x_n\}$ on $[a, b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \delta^2$$

Let $M_i = \sup f(t)$ and $m_i = \inf f(t)$ and $M_i^* = \sup(\phi \circ f(t))$ and $m_i^* = \inf(\phi \circ f(t))$ where the sup, inf values are taken in $t \in [x_{i-1}, x_i]$.

Divides the index set $\{1, \dots, n\}$ as:

$$A = \{i \in \{1, \dots, n\} \mid M_i - m_i < \delta\} \text{ and } B = \{i \in \{1, \dots, n\} \mid M_i - m_i \geq \delta\}.$$

Now, if $i \in A$, then $M_i^* - m_i^* = \sup(\phi \circ f(t)) - \inf(\phi \circ f(t)) \leq \epsilon$

because $\forall s, t \in [x_{i-1}, x_i], |f(s) - f(t)| \leq M_i - m_i \Rightarrow |\phi(f(s)) - \phi(f(t))| < \epsilon$.

And, if $i \in B$, then $M_i^* - m_i^* \leq 2 \cdot \sup_{t \in [a, b]} |\phi(t)|$

We obtain that

$$\delta \cdot \sum_{i \in B} \Delta \alpha_i \leq \sum_{i \in B} [M_i - m_i] \cdot \Delta \alpha_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta \alpha_i = U(f, P, \alpha) - L(f, P, \alpha) < \delta^2.$$

That is $\sum_{i \in B} \Delta \alpha_i < \delta$. Finally,

$$\begin{aligned} \sum_{i=1}^n [M_i^* - m_i^*] \Delta \alpha_i &= \sum_{i \in A} [M_i^* - m_i^*] \Delta \alpha_i + \sum_{i \in B} [M_i^* - m_i^*] \Delta \alpha_i \\ &\leq \epsilon \cdot \sum_{i \in A} \Delta \alpha_i + 2 \cdot \sup_{t \in [m, M]} |\phi(t)| \cdot \sum_{i \in B} \Delta \alpha_i \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a)] + 2 \sup |\phi| \cdot \delta \\ &\leq \epsilon \cdot [\alpha(b) - \alpha(a) + 2 \sup |\phi|] \end{aligned}$$

Proof of Thm 8.1.11.

Suppose that $f \in \mathcal{R}(a)$. Given $\epsilon > 0$, there is a partition $P = \{x_0, \dots, x_n\}$ on $[a, b]$ s.t.
 $U(f, P) - L(f, P) < \epsilon$, since f is continuous, thus $f' \in \mathcal{R}$.

By the Mean-Value Theorem, for each $1 \leq i \leq n$

$$\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \alpha'(t_i)(x_i - x_{i-1}) = \alpha'(t_i) \cdot \Delta x_i \text{ for some } t_i \in [x_{i-1}, x_i]$$

For any $s_i \in [x_{i-1}, x_i]$,

$$\sum_{i=1}^n |\alpha'(s_i) - \alpha'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n [M_i - m_i] \cdot \Delta x_i < \epsilon.$$

$$\begin{aligned} \text{Now, } & \left| \sum_{i=1}^n f(s_i) \cdot \Delta \alpha_i - \sum_{i=1}^n f(s_i) \cdot \alpha'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n f(s_i) \cdot \alpha'(t_i) \cdot \Delta x_i - \sum_{i=1}^n f(s_i) \alpha'(s_i) \cdot \Delta x_i \right| \\ &= \left| \sum_{i=1}^n [\alpha'(t_i) - \alpha'(s_i)] \cdot f(s_i) \cdot \Delta x_i \right| \leq \sup |f| \cdot \epsilon \quad (*) \end{aligned}$$

$$\text{Therefore, } \left| \int_a^b f \, d\alpha - \int_a^b f \alpha' \, dx \right| \leq |U(f, P, \alpha) - U(f \alpha', P, x)| \leq \sup |f| \cdot \epsilon$$

Since (*) holds for any refinement of P .

Proof of Thm 0.1.13.

Enough to show $(1) \Leftrightarrow (2)$. Let $\int_a^b f = A$.

Given $\varepsilon > 0$, there exists $P_0 = \{x_0, x_1, \dots, x_n\}$ s.t. $U(f, P_0) < A + \varepsilon$.

Put $M = \sup |f|$, $\delta_1 = \frac{\varepsilon}{nM}$. Let $P \in \mathcal{P}[a, b]$ s.t. $\|P\| < \delta_1$. Write $P = \{x_0, \dots, x_m\}$.

Let $J = \{i \in \{1, \dots, m\} \mid (x_{i-1}, x_i) \cap P_0 \neq \emptyset\}$ ($M \delta_1, f > 0$. ($f = f + \min f$))

$J = \{j \in \{1, \dots, m\} \mid (x_{j-1}, x_j) \cap P_0 = \emptyset\}$

$$\text{Now, } U(f, P) = \sum_{i=1}^m M_i (x_i - x_{i-1})$$

$$= \sum_{i \in J} M_i (x_i - x_{i-1}) + \sum_{i \notin J} M_i (x_i - x_{i-1})$$

$$< nM \|P\| + U(f, P_0) < A + 2\varepsilon.$$

$\nearrow$ $\sum_{i \in J} M_i (x_i - x_{i-1}) < nM \|P\|$ $\nearrow$ $\sum_{i \notin J} M_i (x_i - x_{i-1}) < U(f, P_0)$

Similarly $L(f, P) > A - 2\varepsilon$, for some $\delta_2 > 0$, $\|P\| < \delta_2$

Thm. Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded map and continuous except finitely many points. Then f is Riemann integrable.

Proof. Define $P_n = [a, b] \cap \left\{ \frac{k}{2^n} \mid k \in \mathbb{Z} \right\}$. Let A be a set of discontinuous points of f .

Put N be a number of elements of A . Now, for any $n \in \mathbb{N}$, there is at most $2N$ for $i \in \mathbb{N}$

$$\text{s.t. } A \cap [x_{i-1}, x_i] \neq \emptyset$$

where $x_i \in P_n$. Given $\varepsilon > 0$, take $n \in \mathbb{N}$ s.t. $2 \cdot \sup |f| \cdot 2N \cdot \frac{1}{2^n} < \frac{\varepsilon}{2}$.

Meanwhile, f is continuous on the set

$$C = \bigcup_{A \cap [x_{i-1}, x_i] = \emptyset} [x_{i-1}, x_i]$$

Therefore there is a partition P' on C s.t. $U(f, P') - L(f, P') < \frac{\varepsilon}{2}$.

Finally, for $P = P_n \cup P'$,

$$U(f, P) - L(f, P) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} < \varepsilon.$$

□

Proof of Lemma 0.1.25.

Let $P_n = \{a, s, s + \frac{1}{n}, b\}$ for large enough $n \in \mathbb{N}$ ($\frac{1}{n} < b - s$).

Then, $\Delta\alpha_1 = \alpha(s) - \alpha(a) = 0$, $\Delta\alpha_2 = 1$, $\Delta\alpha_3 = 0$. Now

$$\inf_{t \in [s, s + \frac{1}{n}]} f(t) \leq \int f d\alpha \leq \int f d\alpha \leq \sup_{t \in [s, s + \frac{1}{n}]} f(t)$$

And since f is continuous at s , $\lim_{n \rightarrow \infty} \inf_{t \in [s, s + \frac{1}{n}]} f(t) = \lim_{n \rightarrow \infty} \sup_{t \in [s, s + \frac{1}{n}]} f(t) = f(s)$.

Proof of Thm 0.1.26.

Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t. $\sum_{n=N+1}^{\infty} C_n < \varepsilon$.

Let $\alpha_1(x) = \sum_{n=1}^N C_n I(x - s_n)$ and $\alpha_2(x) = \sum_{n=N+1}^{\infty} C_n I(x - s_n)$. Then $\alpha = \alpha_1 + \alpha_2$.

First, $\int_a^b f d\alpha_1 = \sum_{n=1}^N C_n \cdot f(s_n)$, and next, for $M = \sup |f|$

$$\left| \int_a^b f d\alpha_2 \right| \leq \int_a^b |f| d\alpha_2 \leq M \cdot \int_a^b 1 d\alpha_2 = M \cdot \sum_{n=N+1}^{\infty} C_n < M\varepsilon.$$

Now, $\int_a^b f d\alpha = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2 = \sum_{n=1}^N C_n f(s_n) + \int_a^b f d\alpha_2$

$$\Rightarrow \left| \int_a^b f d\alpha - \sum_{n=1}^N C_n f(s_n) \right| < M\varepsilon, \text{ and letting } N \rightarrow \infty.$$

Thm. Let $f: [a, b] \rightarrow \mathbb{R}$ be given, TFAE:

- f is bounded variation.
 - There exist monotone increasing $g, h: [a, b] \rightarrow \mathbb{R}$ such that $f = g - h$.
- Proof. Suppose f is bounded variation. Define

$$g(x) = V_a^x(f) \quad \text{and} \quad h(x) = V_a^x(f) - f(x).$$

By definition, g is monotone increasing. And, if $a \leq x < y \leq b$, then

$$h(y) - h(x) = V_a^y(f) - V_a^x(f) - (f(y) - f(x)) = V_x^y(f) - (f(y) - f(x)) \geq 0.$$

The converse is trivial. $\square$

Exercise 6

#1. Suppose α increase on $[a, b]$, and continuous at $x_0 \in [a, b]$.

If $f(x_0) = 1$ and $f(x) = 0$ for all $x \neq x_0$, then $f \in \mathcal{B}(\alpha)$ and $\int f d\alpha = 0$.

Proof. Let $P_n := \{a, x_0 - \frac{1}{n}, x_0, x_0 + \frac{1}{n}, b\} \cap [a, b]$.

Given $\epsilon > 0$, there exists $\delta > 0$ s.t. $|x_0 - t| < \delta \Rightarrow |\alpha(x_0) - \alpha(t)| < \frac{\epsilon}{2}$

Now, for $N \in \mathbb{N}$ s.t. $1 < N\delta$, now,

$$0 = L(f, P_N, \alpha) \leq \int f d\alpha \leq \bar{\int} f d\alpha \leq U(f, P_N, \alpha) < \epsilon.$$

#2. Suppose that for some $x_0 \in [a, b]$, $f(x_0) > 0$.

Then, there exists $\delta > 0$ s.t. $|t - x_0| < \delta \Rightarrow |f(t) - f(x_0)| < \frac{1}{2} \cdot f(x_0)$.

By the triangle inequality $-\frac{1}{2}f(x_0) + |f(x_0)| < f(t) < \frac{1}{2}f(x_0) + |f(x_0)|$, if $|t - x_0| < \delta$.

Let $P \in \mathcal{P}[a, b]$ be any partition. Then, for a refinement

$$P^* = P \cup \{x_0 - \delta, x_0 + \delta\} \cap [a, b],$$

$$\text{Now, } \frac{f(x_0)}{2} \cdot \delta \leq U(f, P^*) \leq U(f, P)$$

But, we can take $P' \in \mathcal{P}[a, b]$ s.t. $U(f, P') < \frac{f(x_0)}{2} \cdot \delta$ since $\int_a^b f = 0$, it is contradiction.

#3. Define $\beta_1, \beta_2, \beta_3$ on $[-1, 1]$ s.t.

$$\beta_3(x) = 0 \text{ if } x < 0, \quad \beta_3(x) = 1 \text{ if } x > 0, \text{ and } \beta_1(0) = 0, \beta_2(0) = 1, \beta_3(0) = \frac{1}{2}.$$

Let f be a bounded map on $[-1, 1]$.

a). $f \in \mathcal{B}(\beta_1)$ if and only if $f(0^+) = f(0)$ and thus then $\int f d\beta_1 = f(0)$

b).

c). $f \in \mathcal{B}(\beta_3)$ if and only if f is continuous at 0.

d). If f is continuous at $x=0$, then

$$\int f d\beta_1 = \int f d\beta_2 = \int f d\beta_3 = 0.$$

Proof. Let $P_n = \{-1, 0, \frac{1}{n}, 1\}$. Then, for each $i=1, 2, 3$,

$$U(f, P_n, \beta_i) = M_1 \cdot [\beta_i(0) - \beta_i(-1)] + M_2 \cdot [\beta_i(\frac{1}{n}) - \beta_i(0)] + M_3 \cdot [\beta_i(1) - \beta_i(\frac{1}{n})]$$

$$= M_1 \beta_i(0) + M_2 [1 - \beta_i(0)] + 0$$

$$= (M_1 - M_2) \beta_i(0) + M_2 \quad \text{Similarly}$$

$$L(f, P_n, \beta_i) = (m_1 - m_2) \beta_i(0) + m_2.$$

When $i=1$, $\beta_1(0)=0$, therefore

$$U(f, P_n, \beta_1) - L(f, P_n, \beta_1) = M_2 - m_2 = \sup_{t \in [0, \frac{1}{n}]} f(t) - \inf_{t \in [\frac{1}{n}, 1]} f(t) \rightarrow 0 \text{ as } n \rightarrow \infty,$$

since $f(0^+) = f(0)$. To show b), take $P_n = \{-1, -\frac{1}{n}, 0, \frac{1}{n}, 1\}$. Then,

$$U(f, P_n, \beta_i) = M_1 [\beta_i(-\frac{1}{n}) - \beta_i(-1)] + M_2 [\beta_i(0) - \beta_i(-\frac{1}{n})] + M_3 [\beta_i(\frac{1}{n}) - \beta_i(0)] + M_4 [\beta_i(1) - \beta_i(\frac{1}{n})]$$

$$= M_2 \beta_i(0) + M_3 (1 - \beta_i(0)). \quad \text{Similarly}$$

$$L(f, P_n, \beta_i) = m_2 \beta_i(0) + m_3 (1 - \beta_i(0))$$

When $i=2$, $\beta_2(0)=1$. Therefore

$$U(f, P_n, \beta_2) - L(f, P_n, \beta_2) = M_2 - m_2 \rightarrow 0 \text{ if } f(0^-) = f(0).$$

When $i=3$, $\beta_3(0)=\frac{1}{2}$, then

$$U(f, P_n, \beta_3) - L(f, P_n, \beta_3) = \frac{1}{2} [M_2 - m_2 + M_3 - m_3] \rightarrow 0 \text{ if } f(0^+) = f(0) = f(0^-).$$

Moreover, $U(f, P_n, \beta_i) \rightarrow f(0) \cdot [\beta_i(0) + (1 - \beta_i(0))] = f(0) \text{ as } n \rightarrow \infty.$

□

#4. Dirichlet's map is Not Riemann integrable.

Proof. Let $P \in \mathcal{P}[a,b]$ be any partition. Then, by the density of Rationals and irrationals

$M_i = 1$, $m_i = 0$ for all i . That is,

$$U(f, P) = 1 \quad \text{and} \quad L(f, P) = 0.$$

Thus, $P \notin \mathcal{R}(f)$ (WLOG. $[a,b] := [0,1]$)

#5. f is bounded, and $f^2 \in \mathcal{R}$, then $f \in \mathcal{R}$? No!

$$f: [0,1] \rightarrow \mathbb{R} : x \mapsto \begin{cases} 1 & x \in \mathbb{Q} \\ -1 & x \notin \mathbb{Q} \end{cases}$$

$f^2 = 1$, thus integrable, but f is not.

Meanwhile, if $f^3 \in \mathcal{R}$, then $f \in \mathcal{R}$ because the real value of $\sqrt[3]{x}$ is unique.

#6. Let C be a Cantor set.

Define the Characteristic function of A as

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Clearly, $\chi_A + \chi_{A^c} = 1$. Now, observe that for each $k \in \mathbb{N}$ and any $P \in \mathcal{P}[a, b]$,

$$U(f, P) - L(f, P) \leq U(f \cdot \chi_{C_k}, P) + U(f \cdot \chi_{C_k^c}, P) - L(f \cdot \chi_{C_k}, P) - L(f \cdot \chi_{C_k^c}, P)$$

Since $C \subseteq C_k$, $C_k^c \subseteq C^c$. Therefore, $f \cdot \chi_{C_k^c}$ is continuous map except finitely many points, thus Riemann-integrable. Meanwhile, for each $m \in \mathbb{N}$, define

$$P_m = \left\{ \frac{a}{3^m} \mid 0 \leq a \leq 3^m, a \in \mathbb{Z} \right\}$$

Then, $U(f \cdot \chi_{C_k}, P_m) - L(f \cdot \chi_{C_k}, P_m) \leq 2 \cdot \sup |f| \cdot 2^k \cdot [3^{m-k} + 2] \cdot \frac{1}{3^m} \rightarrow 0$ as $k, m \rightarrow \infty$

Thus proved. (More Technically, we can take $m = k+1$.)

[General Technic for χ_A .

Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded map, and $A \subset [a, b]$ be a subset.

Clearly, $\chi_A + \chi_{A^c} = 1$. Therefore, $f = f \cdot \chi_A + f \cdot \chi_{A^c}$.

Since $\sup_{x \in A} (f + g) \leq \sup_{x \in A} f + \sup_{x \in A} g$, for any partition P on $[a, b]$,

$$\begin{aligned} U(f, P) &= \sum \sup f \cdot \Delta x_i = \sum \sup (f \chi_A + f \chi_{A^c}) \cdot \Delta x_i \leq \sum \sup f \cdot \chi_A \cdot \Delta x_i + \sum \sup f \cdot \chi_{A^c} \cdot \Delta x_i \\ &= U(f \cdot \chi_A, P) + U(f \cdot \chi_{A^c}, P) \end{aligned}$$

The lower sum too. Therefore,

$$U(f, P) \leq U(f \chi_A, P) + U(f \chi_{A^c}, P) \quad \text{and} \quad L(f, P) \geq L(f \chi_A, P) + L(f \chi_{A^c}, P).$$

suppose that the given f is continuous on $[a, b]$ except a countable subset $C \subset [a, b]$.
Denote $C = \{d_i \mid i \in \mathbb{N}\}$. Given $\varepsilon > 0$, define

$$C_n = \bigcup_{i=1}^n \left[d_i - \frac{\varepsilon}{2^{i+1}}, d_i + \frac{\varepsilon}{2^{i+1}} \right] \cap [a, b].$$

$$\sum_{i=1}^n \frac{1}{2^i} = \frac{1}{2} \cdot \frac{1 - \frac{1}{2^{n+1}}}{1 - \frac{1}{2}} = \frac{2^{n+1} - 1}{2^{n+1}}.$$

#7. Let $f \in \mathcal{R}$ on $[0,1]$, and $f \in \mathcal{R}$ on $[c,1]$ for any $c > 0$.

Define
$$\int_0^1 f(x) dx = \lim_{c \rightarrow 0} \int_c^1 f(x) dx$$

if the limit exists.

a). If $f \in \mathcal{R}$ on $[0,1]$, then this definition equals to old one.

b). Construct f s.t. the limit exists, but f does not.

Proof.

#8. Suppose $f \in \mathcal{R}$ on $[a,b]$ for every $b > a$ where a is fixed. Define

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

if the limit exists. Assume $f(x) \geq 0$ and decreases monotonically on $[1, \infty)$. Prove

$$\int_1^\infty f(x) dx \text{ converges if and only if } \sum_{n=1}^\infty f(n) \text{ converges.}$$

Proof.

#9, Let $b > 0$. Then,

$$\begin{aligned} \int_0^b \frac{\cos x}{1+x} dx &= \frac{\sin x}{1+x} \Big|_0^b + \int_0^b \frac{\sin x}{(1+x)^2} dx \\ &= \frac{\sin b}{1+b} + \int_0^b \frac{\sin x}{(1+x)^2} dx. \end{aligned}$$

Since $\int_0^b \left| \frac{\sin x}{(1+x)^2} \right| dx \leq \int_0^b \frac{1}{(1+x)^2} dx = \left. \frac{-1}{1+x} \right|_0^b = 1 - \frac{1}{1+b} \rightarrow 1$ as $b \rightarrow \infty$,

$\int_0^\infty \frac{\sin x}{(1+x)^2} dx$ exists, so is $\int_0^\infty \frac{\cos x}{1+x} dx$.

But $\int_0^\infty \frac{|\cos x|}{1+x} dx$ does not exist, because

In $n\pi + [-\frac{\pi}{3}, \frac{\pi}{3}]$, $|\cos x| \geq \frac{1}{2}$. Now, define for each $n \in \mathbb{N}$,

$$I_n = \int_{-\frac{\pi}{3} + n\pi}^{\frac{\pi}{3} + n\pi} \frac{|\cos x|}{1+x} dx. \text{ Clearly, } \sum_{n=1}^m I_n \leq \int_0^{\frac{\pi}{3} + m\pi} \frac{|\cos x|}{1+x} dx$$



But $I_n \geq \frac{2\pi}{3} \cdot \frac{1}{2} \cdot \int_{-\frac{\pi}{3} + n\pi}^{\frac{\pi}{3} + n\pi} \frac{1}{1+x} dx$, thus proved.

(Since $\frac{1}{1+x}$ decreasing, $\sum_{n=1}^\infty \int_{-\frac{\pi}{3} + n\pi}^{\frac{\pi}{3} + n\pi} \frac{1}{1+x} dx \geq 2 \int_0^\infty \frac{1}{1+x} dx$.)

#12. Let $f: [a, b] \rightarrow \mathbb{R}$ be given with $f \in \mathcal{B}(a)$.

For any $\varepsilon > 0$, there exists a continuous $g: [a, b] \rightarrow \mathbb{R}$ s.t. $\|f - g\|_2 < \varepsilon$.

Proof. Given $\varepsilon > 0$, take $P = \{x_0, \dots, x_n\} \in \mathcal{P}[a, b]$ s.t.

$$U(f, P, \alpha) - L(f, P, \alpha) < \varepsilon.$$

$$\text{define } g(t) := \frac{x_i - t}{x_i - x_{i-1}} \cdot f(x_{i-1}) + \frac{t - x_{i-1}}{x_i - x_{i-1}} \cdot f(x_i) \quad (x_{i-1} \leq t \leq x_i)$$

Then, g is continuous on $[a, b]$: (x_{i-1}, x_i) , clearly continuous and

$$g(x_{i-1}) = g(x_i) = g(x_{i-1}^+)$$

Moreover, $|g(t) - f(t)| \leq M_i - m_i$ for any $1 \leq i \leq n$, therefore

$$|g(t) - f(t)| \cdot \Delta x_i \leq [M_i - m_i] \cdot \Delta x_i.$$

$$\|g - f\|_2 = \left[\sum_{i=1}^n \int_{x_{i-1}}^{x_i} |g - f|^2 dx \right]^{\frac{1}{2}}$$

#13. Define $f(x) = \int_x^{x+1} \sin(t^2) dt$.

a) Prove that for $x > 0$, $|f(x)| < \frac{1}{x}$.

b) Prove that $2xf(x) = \cos(x^2) - \cos[(x+1)^2] + r(x)$ where $|r(x)| < \frac{c}{x^2}$ for some $c \in \mathbb{R}$.

c) Evaluate $\limsup_{x \rightarrow \infty} xf(x)$ and $\liminf_{x \rightarrow \infty} xf(x)$.

d) Does $\int_0^\infty \sin(t^2) dt$ Converge?

Proof. Put $t^2 = u$. Then $2t dt = du$,

$$\begin{aligned} f(x) &= \int_x^{x+1} \sin(t^2) dt = \int_{x^2}^{(x+1)^2} \sin u \cdot \frac{1}{2\sqrt{u}} du \\ &= -\frac{\cos u}{2\sqrt{u}} \Big|_{x^2}^{(x+1)^2} - \int_{x^2}^{(x+1)^2} \cos u \cdot \frac{1}{4u^{\frac{3}{2}}} du \\ &= \frac{\cos x^2}{2x} - \frac{\cos(x+1)^2}{2(x+1)} - \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}} du \end{aligned}$$

$$\begin{aligned} \text{Now, } |f(x)| &\leq \frac{1}{2x} + \frac{1}{2(x+1)} + \int_{x^2}^{(x+1)^2} \frac{1}{4u^{\frac{3}{2}}} du \\ &= \frac{1}{2x} + \frac{1}{2(x+1)} + \left(-\frac{1}{2}\right) \cdot \left[u^{-\frac{1}{2}} \Big|_{x^2}^{(x+1)^2} \right] \\ &= \frac{1}{2x} + \frac{1}{2(x+1)} - \frac{1}{2} \left[\frac{1}{x+1} - \frac{1}{x} \right] \\ &= \frac{1}{x} \end{aligned}$$

$$\text{Meanwhile, } 2xf(x) = \cos x^2 - \cos(x+1)^2 + \cos(x+1)^2 \cdot \frac{1}{2(x+1)} - 2x \cdot \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}} du$$

observe that

$$\begin{aligned} \left| \cos(x+1)^2 \cdot \frac{1}{2(x+1)} - 2x \cdot \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}} du \right| &\leq \frac{1}{2} \cdot \left| \frac{1}{x+1} \right| + 2|x| \cdot \int_{x^2}^{(x+1)^2} \frac{1}{4u^{\frac{3}{2}}} du \\ &\leq \frac{1}{2|x|} + \frac{1}{|x+1|} \leq \frac{3}{2x}. \end{aligned}$$

Clearly, $\limsup_{x \rightarrow \infty} r(x) = \liminf_{x \rightarrow \infty} r(x) = 0$. And,

$$\cos x^2 - \cos(x+1)^2 =$$

Clearly, $\limsup_{x \rightarrow \infty} r(x) = \liminf_{x \rightarrow \infty} r(x) = 0$. And, to evaluate upper and lower limit of the remain, take $x_n = 2n \cdot \sqrt{2}$ and $y_n = (2n+1)\sqrt{2}$. Now,

$$\cos x_n^2 - \cos(x_{n+1})^2 = 1 - \cos(2n\sqrt{2} + 1)$$

$$\cos y_n^2 - \cos(y_{n+1})^2 = -1 + \cos((2n+1)\sqrt{2} + 1)$$

Since $\{\cos(n\alpha) \mid n \in \mathbb{N}\}$ is dense in $[-1, 1]$,

$$\limsup_{x \rightarrow \infty} [\cos x^2 - \cos(x+1)^2] = 2 \quad \text{and} \quad \liminf_{x \rightarrow \infty} [\cos x^2 - \cos(x+1)^2] = -2.$$

Finally, let's determine whether $\int_0^\infty \sin t^2 dt$ Converges or diverges.

$$\text{Since } \int_0^\infty \sin t^2 dt = \int_0^\infty t \cdot \sin t^2 \cdot \frac{1}{t} dt$$

$$\sum_{k=0}^{\infty} k f(k) \cdot \frac{1}{k} \quad \text{exists,}$$

But, if $\lim_{N \rightarrow \infty} \sum_{k=0}^{N-1} f(k)$ exists, then is,

$$\text{Since } \int_0^N \sin t^2 dt = \sum_{k=0}^{N-1} \int_k^{k+1} \sin t^2 dt = \sum_{k=0}^{N-1} f(k)$$

Using the Dirichlet's Test, then

$$\int_0^\infty \sin t^2 dt = \sum_{k=0}^{\infty} f(k) = \sum_{k=0}^{\infty} k f(k) \cdot \frac{1}{k}$$

Converges

#15. Suppose f is real, continuously differentiable, $f(a)=f(b)=0$, with

$$\int_a^b (f \circ f)'(x) dx = 1.$$

Prove that $\int_a^b x f(x) f'(x) dx = -\frac{1}{2}$ and $\int_a^b [f'(x)]^2 dx \cdot \int_a^b x^2 f(x) dx \geq \frac{1}{4}$.

Proof. $\int_a^b f^2(x) dx = x f^2(x) \Big|_a^b - \int_a^b x \cdot 2f(x) f'(x) dx = 1$

Since the first term on the right is zero,

$$\int_a^b x \cdot f(x) f'(x) dx = -\frac{1}{2}.$$

Now, $\int_a^b [f'(x)]^2 dx \cdot \int_a^b [x f(x)]^2 dx \geq \left| \int_a^b x f(x) f'(x) dx \right|^2 = \frac{1}{4}$

By Hölder's inequality.

Prop. Let $f: [a, b] \rightarrow \mathbb{R}$ be a differentiable map.

If f is bounded variation, then

$$\int_a^b |f'(x)| dx \leq TV(f)$$

Proof. For any partition $P = \{x_0, \dots, x_n\}$ on $[a, b]$, the Mean-Value Theorem gives that

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \rightarrow \int_a^b |f'(x)| dx \quad \text{as } \|P\| \rightarrow 0.$$

Thus,

